

FieldComms is a self-contained emergency communications server built on a Raspberry Pi 5 for McHenry County RACES, ARES, and Starcom operations. Any smartphone, tablet, or laptop connects to the **EMCOMM-NET** Wi-Fi access point and reaches the full dashboard at <http://192.168.50.1> — no internet, no app installation, no per-device configuration required. All 30 tools, all ICS forms, all reference materials, and all offline maps remain fully functional when the site has no internet connectivity.

■ AMATEUR RADIO	■ STARCOM / PUBLIC SAFETY	■ ICS INCIDENT COMMAND
• Net Control Logger — FCC auto-fill • Callsign Lookup — 800K licensees offline • Observer Mode — read-only agency view • APRS Tactical Map — Graywolf + YAAC • HF Propagation — band conditions live • Repeater Database — ARES/band filters • Dead Man's Switch — net inactivity alert • NTS Radiogram — ARRL formatted traffic • JS8Call — HF keyboard digital messaging • Pat Winlink — backup email over radio	• Starcom Net Logger — Radio ID / unit-based • Weather Net — storm spotter check-in • SAR Net — search and rescue radio ops • Resource Tracking Map — unit placement • Resource Board — 5-state status cycle • Member Roster — certifications, activations • Facilities Directory — EOC, shelters, hospitals • Multiple simultaneous nets on one screen • Observer link for served agencies • ICS platform integration from Starcom mode	• Command — objectives, safety, staff • Operations — T-card board, assignments • Planning — IAP tracker, resource status • Logistics — ICS-205 comms plan, supplies • Finance/Admin — cost, time, procurement • Planning P — 15-phase planning cycle guide • ICS-213 / ICS-214 / ICS-309 forms • Winlink Form Import — XML to incident • Incident types: Disaster, HazMat, SAR, MCI • Multi-user — all sections see live data

■ Connectivity — Dual WAN with Automatic Failover

InstyConnect Cellular (Primary WAN)	Starlink Satellite (Automatic Failover)	AMPRNet / 44Net (Amateur Radio IP)
InstyConnect Drum omnidirectional antenna + Switchblade directional backup. T-Mobile + Verizon dual-carrier. Pauses at \$5/month between activations.	Automatic failover when cellular drops. ASUS RT-BE58 Go manages failover with no operator action required. All FieldComms features remain active during satellite operation.	Dedicated Raspberry Pi 5 gateway. WireGuard tunnel to amprgw.ampr.org. Routes 44.0.0.0/8 for all EMMCOMM-NET devices. AMPRNet allocation required from portal.ampr.org.

■ Hardware Platform

COMPONENT	SPECIFICATION	ROLE
FieldComms Server	Pi 5 16 GB · Pironman MAX 5 · 2× 1 TB NVMe RAID 1	32 pages · 15 services · FCC DB · Kiwix · maps · Pat
44Net Gateway	Pi 5 16 GB · Argon NEO 5 · 256 GB SSD · Pi OS Desktop	WireGuard to AMPRNet · routes 44.0.0.0/8 · status at :9000
Wi-Fi + Switch	ASUS RT-BE58 Go Wi-Fi 7 · UniFi Lite 16 PoE (16-port GbE)	EMCOMM-NET · DHCP · dual WAN failover · wired hub
Primary WAN (Cellular)	InstyConnect Drum (omni) + Switchblade (directional) · 5G/LTE	T-Mobile + Verizon · pause at \$5/month · swap in 5 min
Secondary WAN (Satellite)	Starlink dish + Ethernet adapter → ASUS USB WAN	Auto-failover when cellular drops · all features stay active
Radio + Workstations	IC-7300 · Winlink · VARA HF · JS8Call · Pi 500 ×4 · Monitor ×4	HF comms · browser operator stations · no app install

Reference & Administration

■ Kiwix Library	WikiMed, Wikipedia, iFixit — offline
■ Reference Library	SOGs, plans, field docs — searchable
■ Print Center	ICS 213/214/309, NTS, access cards
■ Cheat Sheets	Phonetics, Q-codes, CTCSS, band plan
■ Callsign Lookup	800K licensees — instant offline
■ Pre-Flight Check	GO / CAUTION / NO-GO readiness
■ WAN Dashboard	InstyConnect, Starlink, 44Net live
■ Offline Maps	USGS topo — APRS and SAR maps
■ Health Monitor	CPU, disk, GPS, internet, services
■ Access Cards	Avery 5371 printable wallet cards

HOW TO ACCESS

- 1 Wi-Fi: **EMCOMM-NET**
- 2 Browser: <http://192.168.50.1>
- 3 Select mode: Amateur / Starcom / ICS

No app · No login · Any device · Any OS

32

Web Pages

15

Services

6

WAN Sources

WHY FIELDCOMMS

Fully Offline	All 32 tools work with zero internet
Multi-User	Every operator sees live data simultaneously
30-Min Deploy	Unpack and operational in under 30 minutes
Dual WAN	Cellular primary, satellite auto-failover
RAID 1	Drive failure does not stop operations
Full ICS	All 5 sections, Planning P, printable IAP
Radio Integrated	Winlink, JS8Call, APRS, PACTOR, Pat
Grant Eligible	FEMA BRIC, ARPA-E, ARRL Foundation, DHS

501(c)(3) Grant Eligible · Funding Sources: FEMA BRIC · ARPA-E · ARRL Foundation · ARES Special Fund · DHS BRIC Nonprofit Security Grant Program · Illinois Homeland Security